

== README FOR ==

Climate Hazards Group Infrared Precipitation with Stations (CHIRPS).
Quasi-global satellite and observation based precipitation estimates over land, 1981 to near-real time.

== PRODUCED BY ==

Climate Hazards Group
Department of Geography
University of California at Santa Barbara

In collaboration with:
The USGS Famine Early Warning Systems Network (FEWS NET)

== SUMMARY ==

The Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) data archive is a quasi-global (50S-50N), gridded 0.05 degree resolution, 1981 to near-real time precipitation time series. The terrestrial precipitation estimates, are available in daily to annual time intervals. In addition to the quasi-global extent, subsets of Western Hemisphere, Africa and the Central America/Caribbean regions are available. These are available in several formats (NetCDF, TIFF, BIL, PNG) and are located in separate subdirectories.

Two CHIRPS products are produced operationally: a rapid preliminary version, and a later final version. The preliminary CHIRPS product is available, for the entire domain, two days after the end of a pentad (2nd, 7th, 12th, 17th, 22nd and 27th). The preliminary CHIRPS uses only Global Telecommunications System (GTS) and Conagua (Mexico) data. The final CHIRPS product takes advantage of several other stations sources and is complete sometime in the third week of the following month. Final products for all times/domains/formats are calculated at that time.

== ONLINE INFO ==

CHIRPS homepage
<http://data.chc.ucsb.edu/data/chirps>

CHIRPS FAQ
http://chg-wiki.geog.ucsb.edu/wiki/CHIRPS_FAQ

CHIRPS paper
<http://pubs.usgs.gov/ds/832/>

== VERSION HISTORY ==

Version 2.0 released 2015.02.12
Version 1.8 released 2014.05.14
Version 1.7 released 2013.05.01

== Changes in Version 2.0 ==

Since v1.8 we have added new stations around the world. In particular, 400 in Peru, 1,200 across Africa from Sharon Nicholson, 11,000 in Brazil, over 1,000 in

Mexico (Conagua), 76 from Southern Africa, 500 in Russia, 250 in Central Asia and 400 across Central America. In addition we have topped off data from several African countries. The Conagua stations are available in near-real time and will directly improve our preliminary CHIRPS product.

Daily data now available for the Globe

NetCDF format added

Replaced erroneous stations in Tanzania

Expanded diagnostics, stations by Country plots, EWX rchecks, station comparison/*badness* plots, station density maps, pentad and monthly png's of excluded stations.

== ACCESS ==

Online display and access,
<http://earlywarning.usgs.gov/fews/ewxindex.php>

Via anonymous ftp,
<ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/>

In a browser,
<ftp://ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/>

Using wget (Linux, Mac, cygwin)
(example to get all global monthly CHIRPS tif's for 2013)
`wget ftp://ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/global_monthly/tifs/chirps-v2.0.2013*`

(In Progress) Google Earth Engine <https://earthengine.google.org>

(In Progress) Snippets: Ability to embed latest CHIRPS images into your webpage

== FOLDERS ==

africa_6-hourly Special binary format.	Africa subset 6-hourly fields at 0.1 degree resolution.
africa_daily resolution. BIL's and TIFF's.	Africa subset daily fields at 0.05 and 0.25 degree
africa_dekad	Africa subset dekadal fields. BIL's, PNG's and TIFF's.
africa_monthly	Africa subset monthly fields. BIL's, PNG's and TIFF's.
africa_pentad	Africa subset monthly fields. BIL's, PNG's and TIFF's.
camer-carib_dekad and TIFF's.	Central America/Caribbean subset dekadal fields. BIL's
camer-carib_monthly and TIFF's.	Central America/Caribbean subset monthly fields. BIL's
camer-carib_pentad and TIFF's.	Central America/Caribbean subset pentadal fields. BIL's
diagnostics processing and results	Various data/plots giving more detail about CHIRPS
docs	CHIRPS related documents.
EAC_monthly	East African Community monthly fields. BIL's and TIFF's.

EAC_monthly_EWX	East African Community monthly EWX fields. TIFF's.
global_2-monthly	Quasi-global 2-monthly fields. TIFF's.
global_2_EWX-monthly	Quasi-global 2-monthly EWX fields. TIFF's.
global_3-monthly	Quasi-global 3-monthly fields. TIFF's.
global_3_EWX-monthly	Quasi-global 3-monthly EWX fields. TIFF's.
global_annual	Quasi-global annual fields. NetCDF's and TIFF's.
global_daily	Quasi-global daily fields. NetCDF's and TIFF's.
global_dekad	Quasi-global dekadal fields. BIL's, NetCDF's and TIFF's.
global_dekad_EWX	Quasi-global dekadal fields. TIFF's.
global_monthly	Quasi-global monthly fields. BIL's, NetCDF's and TIFF's.
global_monthly_EWX	Quasi-global monthly fields. TIFF's.
global_pentad	Quasi-global pentadal fields. BIL's, NetCDF's and TIFF's.
prelim	Quasi-global Rapid CHIRPS with 2-day lag. GTS and Conagua (Mexico) stations only.
whem_daily	Western Hemisphere daily fields. TIFF's.

EWX: EWX files are data, anomaly and zscore used to populate the Early Warning eXplorer

@EROS (operational):

<http://earlywarning.usgs.gov/fews/ewxindex.php>

@CHG (development/experimental): <http://chg-ewx.geog.ucsb.edu:8080/EWX/index.html>

== DATA SET CITATION ==

Funk, C.C., Peterson, P.J., Landsfeld, M.F., Pedreros, D.H., Verdin, J.P., Rowland, J.D., Romero, B.E., Husak, G.J., Michaelsen, J.C., and Verdin, A.P., 2014, A quasi-global precipitation time series for drought monitoring: U.S. Geological Survey Data Series 832, 4 p.
<ftp://ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/docs/USGS-DS832.CHIRPS.pdf>

== PUBLICATIONS ==

<ftp://ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/docs/USGS-DS832.CHIRPS.pdf>

== GET ADDED TO OUR USERS LIST ==

If you would like to receive updates on CHIRPS processing/validations/publications etc. send an email to chirps@geog.ucsb.edu

== CONTACT ==

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== UPDATES/MESSAGES ==

April 2015:

CHIRPS data has recently been reprocessed for 1) CHIRPS final March 2015 files (complete replacement) and 2) daily CHIRPS from 2000 to present. If you are not interested in either of these two products, you can ignore the rest of this message.

In summary, two updates to CHIRPS 2.0 have been made. First, CHIRPS final data for March 2015 (all time steps/domains/formats) have been updated with certain stations removed due to a new irregularity in the stations. Second, daily (not pentads, dekads or months) CHIRPS data from 2000-present have been updated to correct areas falling in gaps in the IR coverage. Outside of those gap areas the daily CHIRPS are not affected.

1: CHIRPS final

After posting the final CHIRPS for March 2015, we found some locations where the station data was obviously wrong. We have since developed an extra screening step for catching this type of problem and reprocessed all the CHIRPS final data for March 2015. If you downloaded March data earlier this week (before Thursday, March 23rd), you should go back and download the data again. This new screening algorithm process will be included in our processing going forward.

2: CHIRPS daily

Thanks to feedback from Marcel Kuettel, we realized a problem in daily CHIRPS for the CPC-IR time period, 2000 to present. There was a problem creating the daily percent cold cloud duration (%CCD) map used in the downscaling. Places with missing IR data were set to zero instead of missing. Anywhere this happened, the precipitation was set to zero. This was always a problem for Eastern Australia/Indonesia/Japan, where a gap between two geostationary satellites exists. There were a small number of times this problem occurred outside of these particular areas. Browse .png images of daily %CCD before the fix:

ftp://ftp.chc.ucsb/pub/org/chg/products/CHIRPS-2.0/diagnostics/CPC-IR_perctCCD_pngs/

under the subdirectories

all_days_by_month

singleProblemDays_outsideAustralia

If your region of interest falls within the negative (yellow) areas, you should re-acquire the daily data for that entire month as problems with a given day effect the partitioning of precipitation among the other days in that month.

/end April 2015 message

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